

Breathe: Communicating Trust in Public Air Quality Data

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Introduction

Air quality numbers are easy to read but harder to trust than they first appear. A runner may check the Air Quality Index before heading outside. A parent may look at a pollution forecast before letting a child play outdoors. Someone with asthma may check a public dashboard before deciding whether to limit time outside. In each case, the number on the screen is not just information. It can shape a real decision.

Most public air quality dashboards describe the air quality. Much fewer explain the state of the data behind that reading. A dashboard might show that the air is good, moderate, or unhealthy, but it might not indicate whether the sensor data is recent, complete, consistent, or affected by unusual reporting behavior. That leaves a silent but significant gap: a user might understand what AQI means but still have no obvious way to judge how much trust to put in the data that generated it.

That gap matters because environmental dashboards depend on data pipelines, and data pipelines fail in ordinary ways. Sensors stop reporting. Readings go out of date. Nearby monitors disagree. A device may repeat the same value for too long, or a sudden spike may appear without enough context to know whether it reflects a real environmental event or a data quality problem. These scenarios don't inherently invalidate the air quality information, but they do mean the user deserves additional context before treating the reading as fully reliable.

In this paper, we present Breathe, a trust-aware environmental data dashboard to investigate how public air quality interfaces can communicate both environmental risk and data trustworthiness. Breathe distinguishes between two questions that are often confused: What is the current air quality? How trustworthy does the sensor data behind it seem to be? The first

question is answered through AQI status and plain-language recommendations. The second is answered through data freshness, completeness, consistency, anomaly evidence, and an explainable trust score.

Breathe is not intended to replace official systems such as EPA, AirNow, OpenAQ, Mecklenburg County Air Quality, or NC DEQ resources. Instead, it is a research prototype that adds a user-facing trust layer on top of environmental data. The goal is not to declare whether official readings are correct or incorrect. Rather, the goal is to help users understand when the available data appears reliable, when it deserves caution, and why that distinction matters.

The guiding research question for this work is, "How can public environmental dashboards communicate air quality risk while also helping everyday users understand the trustworthiness of the underlying sensor data?" Breathe approaches that question at the intersection of environmental informatics, sensor data quality, cybersecurity, and human-centered dashboard design. Its central argument is simple: public dashboards should not only show what the data says. They should also help users understand how much confidence the data deserves.

Background and Motivation

Air quality data is often treated as a simple public information service: collect a number, place it into a category, and display it on a website or mobile app. For many users, though, that number becomes the basis for action. If the air appears safe, they may run, work outdoors, or send children outside. If it appears unhealthy, they may stay indoors, wear a mask, or change their plans. In that sense, air quality dashboards are not just technical interfaces. They are public-facing systems that can shape health, safety, and trust.

That creates an ethical responsibility in how environmental data is presented. A dashboard does not need to be intentionally misleading to confuse people. Even when data is

collected in good faith, users can still misunderstand its reliability if the interface only shows the final reading. If a sensor has stopped reporting, if a reading is delayed, or if nearby sensors disagree, the user may have no practical way to see that uncertainty. The issue is not only whether the data is accurate. It is whether the system explains its limits clearly enough for people to make informed decisions.

This concern is especially important because air quality information exists across different types of data systems. Official historical datasets, such as EPA AirData and AQS, are valuable because they support long-term analysis and regulatory work. Real-time public systems, such as AirNow or local air quality dashboards, are valuable because they help people understand current conditions. Open sensor platforms, such as OpenAQ, are valuable because they provide access to more granular environmental measurements. Each of these sources has a different role, and each comes with different assumptions about timeliness, validation, completeness, and coverage.

The tension between timeliness and certainty is central to this project. Real-time data is useful because people need current information, but current information can also be preliminary. Validated data matters because it has gone through stronger review, but it may arrive too late for daily decisions. Sensor-level data can provide detail, but it can also raise questions about missing values, calibration, reporting gaps, or disagreement between monitors. Breathe is motivated by that tension: how can a dashboard remain useful in the moment while still making the quality of the underlying data visible?

The motivation for Breathe also comes from a broader computing concern. Technical systems increasingly shape public understanding, but users often cannot inspect how those systems work. A person may see a clean air quality card, but behind it are sensors, timestamps,

source systems, normalization rules, update intervals, and assumptions. When those layers are hidden, trust becomes passive. The user is asked to accept the final result without knowing whether the data pipeline appears healthy.

Breathe approaches this problem by treating data trust as part of the user experience. Instead of only asking, "What is the air quality right now?" the system also asks, "Does the data behind this reading appear fresh, complete, and consistent?" This does not mean the system can prove whether a reading is correct. Rather, it means the system can identify basic warning signs that should affect how confidently a user interprets the reading. Missing data, stale readings, sudden spikes, flatlines, and sensor disagreement are not just backend technical details. They are signals that may matter to the people relying on the dashboard.

This research is motivated by the belief that public technology should communicate uncertainty honestly. In the same way that cybersecurity is not only a technical concern but also an ethical responsibility, environmental data systems carry a responsibility to present information in a way that supports informed public decision-making. Breathe is an attempt to explore that responsibility through a practical prototype: a dashboard that communicates both environmental risk and the trustworthiness of the data behind it.

Research Question

The central research question guiding this project is "How can public environmental dashboards communicate air quality risk while also helping everyday users understand the trustworthiness of the underlying sensor data?"

This question separates two ideas that are often presented as one. Air quality risk asks what the current environmental condition may mean for a person. Data trust asks whether the readings behind that condition appear fresh, complete, consistent, and free from obvious

anomaly patterns. Breathe is built around the idea that both questions matter. A moderate AQI reading, for example, may still be incomplete if the interface does not explain whether the supporting data appears reliable.

Several smaller questions follow from the main one. Which sensor data issues matter most for everyday interpretation? Can missing readings, stale data, sudden changes, flatlines, and sensor disagreement be explained without overwhelming the interface? Can a simple trust score help users notice uncertainty without making the system look more authoritative than it is? Finally, can a dashboard communicate caution while still feeling useful, calm, and understandable?

The scope of this research is intentionally limited. Breathe does not attempt to prove that an air quality reading is objectively correct or incorrect. It also does not replace official regulatory systems or environmental agencies. Instead, the project investigates whether a user-facing trust layer can make environmental data quality more visible. The goal is to help users reason more carefully about the data they are already using to make daily decisions, while preserving a clear boundary between public information, data-confidence signals, and official health or regulatory guidance.

Related Work

Breathe builds on several areas of prior work: public air quality monitoring systems, sensor data quality, low-cost air quality sensor reliability, anomaly detection in sensor networks, and trust in data visualization. These areas are usually discussed separately, but Breathe brings them together around one central concern: how ordinary users interpret environmental data when they cannot directly inspect the sensors and data pipelines behind it.

Public air quality monitoring systems already provide important information to communities. In the United States, the EPA's Air Quality System, or AQS, serves as a major repository for ambient air pollution data collected by federal, state, local, and tribal agencies. AQS supports air quality assessments, regulatory designations, modeling, and reporting, and it also contains monitoring station metadata and quality assurance information (United States Environmental Protection Agency, n.d.). This makes AQS valuable as a historical and regulatory reference point. However, the same strength that makes validated data useful for official purposes can also make it less immediate for everyday decisions. People often need to know what the air quality is right now, not months later, after a full validation process.

Real-time public systems help address that need. AirNow provides public-facing AQI information based on data from state, local, and tribal monitoring agencies. However, AirNow also explains that its data receives preliminary quality checks but is not subjected to the full validation process used for EPA's regulatory AQS database (AirNow, n.d.). This distinction is important for Breathe because it shows that real-time environmental data can be useful and preliminary at the same time. The issue is not that preliminary data should be ignored. Rather, users should be given enough context to understand that timeliness and regulatory validation are not the same thing.

Open environmental data platforms add another layer to this landscape. OpenAQ provides open access to ground-level air quality measurements from sources around the world and emphasizes fine-grained physical measurements rather than only aggregated AQI values (OpenAQ, n.d.). This kind of data is useful for research because it can expose individual pollutant measurements, timestamps, locations, and source details. At the same time, open data platforms can vary in coverage and source availability. For Breathe, this supports the need for a

normalized data layer and a trust-aware interface that can communicate source limitations without overwhelming users.

The broader sensor data quality literature also supports the need for this project. Teh, Kempa-Liehr, and Wang (2020) explain that sensor data quality is essential in Internet of Things applications because poor data quality can make the resulting system unreliable for decision-making. Their review identifies common sensor data problems such as missing data, faults, outliers, bias, and drift. This directly connects to Breathe's anomaly detection approach. Missing readings, stale readings, sudden spikes, flatlines, and sensor disagreement are not random features added for effect; they represent practical data quality problems that can influence how much confidence a user should place in a dashboard.

Research on low-cost air quality sensors adds another reason to communicate data trust. Karagulian et al. (2019) review the performance of low-cost sensors for air quality monitoring and explain that these sensors can expand monitoring coverage, but their performance depends on factors such as pollutant type, calibration, environmental conditions, and evaluation method. This does not mean low-cost sensors are not useful. In fact, they can make air quality monitoring more accessible and geographically detailed. However, their readings should be interpreted carefully. Breathe responds to this challenge by making trust and data quality part of the interface rather than hiding them behind the final AQI value.

Prior work on outlier detection in urban air quality sensor networks is also relevant. van Zoest, Stein, and Hoek (2018) note that low-cost urban air quality sensors are more prone to erroneous data than conventional monitors and may produce outliers. Their work also shows that air quality outlier detection is complicated because pollution levels can vary across space and time. This matters for Breathe because a sudden spike is not automatically wrong, and

disagreement between sensors is not always simple to interpret. Breathe therefore uses anomaly detection as a caution signal, not as a final judgment. The system does not claim that an unusual value is false; it communicates that the pattern deserves attention.

Finally, Breathe connects to research on trust in visualization. Elhamdadi et al. (2024) argue that trust plays an important role in how users interpret visualized data and make decisions from it. This is especially important for public dashboards because users often make judgments quickly based on a small number of visual signals. A clean chart or simple AQI card can feel authoritative even when the underlying data has limitations. Breathe is designed around the idea that trust should be communicated directly, not assumed. By showing a trust score, anomaly explanations, and a separation between Daily Mode and Research Mode, the project explores how environmental dashboards can remain simple while still being honest about uncertainty.

Together, this prior work points to a clear gap. Official systems provide validated data, real-time systems provide timely public information, open platforms provide sensor-level access, and research shows that both sensor data quality and visualization trust matter. Dashboard design research also shows that visualization choices shape how people interpret analytic information and make decisions (Hjelle et al., 2024). Yet everyday dashboards still often show air quality readings without clearly explaining freshness, completeness, consistency, source limitations, or anomaly status. Breathe responds to that gap by treating data quality as part of public environmental communication.

System Design

Breathe was designed as a trust-aware environmental telemetry system, not just a conventional air quality dashboard. A conventional dashboard usually emphasizes the final reading: the current AQI, the category, and a short health or activity recommendation. Breathe

keeps those pieces because they are useful, but adds a second layer that evaluates the condition of the data behind the reading. That second layer is the main design move of the project.

The system is implemented as a full-stack web application with a React/Vite frontend and a FastAPI backend. The frontend is responsible for presenting the user interface, managing selected location state, displaying source provenance, and separating simple daily guidance from deeper research information. The backend is responsible for source selection, data ingestion, normalization, anomaly detection, trust scoring, report generation, and API responses. This separation allows the system to keep the user interface clean while placing the more technical reasoning in backend services that can be tested, extended, and documented.

At a high level, Breathe follows a pipeline architecture. A user selects a location or uses the default Charlotte, North Carolina view. The frontend sends the selected location parameters to the backend, and the backend resolves the active source mode. In the current implementation, `'BREATHE_ACTIVE_SOURCE=auto'` attempts to use configured real-time public sources first. OpenAQ serves as the primary source for pollutant-level telemetry; AirNow is supported as an optional second source for public AQI data; EPA AirData/AQS is treated as a historical reference source, and the local fixture remains available only for development, testing, controlled simulations, and explicit fallback. This structure prevents the system from silently treating sample data as real environmental evidence.

After source selection, Breathe converts incoming data into a normalized telemetry model. This normalization step is important because different environmental data sources may represent readings differently. Some sources provide pollutant concentrations, some provide AQI values, some include monitor metadata, and some may provide only limited recent history. By converting readings into a common internal structure, Breathe can apply the same downstream

trust and anomaly logic across different source types. The normalized readings preserve fields such as source identity, pollutant type, value, unit, timestamp, sensor or source identifier when available, location information, data classification, simulation status, and source limitations.

Once readings are normalized, the backend constructs an operational dataset for the current location. This dataset is then passed through the anomaly detection service. The anomaly layer checks for data quality signals such as stale readings, missing readings, sudden spikes or drops, flatline behavior, disagreement between comparable sensors or sources, and limited evidence conditions. The system intentionally treats these signals as indicators for interpretation rather than as proof that a reading is false. For example, a sudden increase may represent a real environmental event, while sparse recent history may simply mean the public source does not provide enough comparable readings for a stronger trend analysis.

The trust scoring service converts those data quality signals into an explainable score from 0 to 100. The purpose of the score is not to determine whether the air is safe. Instead, it estimates whether the available data appears fresh, complete, consistent, and supported by enough evidence. This distinction is essential to the system design. Breathe separates environmental risk from data confidence so that users do not confuse a clean interface or a single AQI value with certainty. A high trust score indicates stronger confidence in the telemetry evidence available to Breathe, while a lower score indicates that the reading should be interpreted with additional caution or compared against official sources.

The frontend is organized around two levels of use. The everyday-facing layer, represented by the Today, Act, Trust, and Learn views, is designed for quick interpretation, cautious action framing, source literacy, and plain-language trust communication. It shows the current AQI or estimated AQI, category, primary pollutant, trust score, source badge, and

plain-language recommendation. Research Mode exposes the technical evidence behind the dashboard. It includes source audit information, anomaly explanations, simulation controls, normalized data traces, report generation, and citation-style exports. This two-mode design allows Breathe to remain accessible to non-technical users while still supporting technical review and research transparency.

Breathe also includes simulation and report-generation features. The simulation controls allow controlled data quality scenarios such as missing readings, stale readings, sudden spikes, flatlines, sensor disagreement, replayed values, and fake-normal behavior. These scenarios are not presented as real-world tampering. Instead, they are controlled research artifacts used to evaluate whether the anomaly and trust scoring logic respond as expected. The report-generation feature creates a structured research snapshot that includes source metadata, AQI basis, trust score, anomaly evidence, limitations, and dataset trace information. This makes the system more useful as a research prototype because it can produce artifacts that support evaluation, documentation, and demo review.

From a deployment perspective, Breathe is implemented as a live AWS-hosted research prototype. The current deployment runs on an Ubuntu EC2 instance with Nginx serving the React/Vite static frontend and reverse-proxying API traffic to a FastAPI backend running privately on 127.0.0.1:8000 under a systemd service. HTTPS is provided through Certbot and Let's Encrypt, and deployment is automated through GitHub Actions over SSH using a server-side deployment script. API keys and environment-specific configuration remain server-side and are not exposed to the frontend bundle. This deployment is intentionally modest rather than highly available, but it demonstrates that the system can operate as a public,

reviewable full-stack prototype with source-aware live telemetry, repeatable builds, and documented operational constraints.

Overall, the system design reflects the project's core argument: environmental dashboards should not only report environmental conditions; they should also communicate the reliability of the data used to produce those conditions. Breathe accomplishes this by combining modular ingestion, normalized telemetry, source provenance, anomaly detection, explainable scoring, and separate user-facing and research-facing views. This architecture makes the project both a practical dashboard and a computer science research prototype for trustworthy environmental telemetry.

Data Sources and Data Ethics

Breathe uses a source-aware data strategy because not all environmental data means the same thing. Real-time preliminary data, validated historical data, development fixtures, and controlled simulations each serve different purposes. If those categories are blurred, a dashboard can mislead users even when the displayed number is technically correct. For that reason, source labeling is treated as an ethical requirement, not just a backend detail.

OpenAQ serves as the primary real public telemetry source in the current prototype. It provides pollutant-level measurements that Breathe can normalize, analyze, and present with provenance metadata. When upstream AQI is unavailable, Breathe may estimate AQI from pollutant readings, but the interface labels that value as estimated rather than source-provided. AirNow is supported as an optional second source for public AQI data when configured or explicitly selected. When AirNow data is used, Breathe treats it as useful real-time public information, but still distinguishes preliminary current data from fully validated regulatory evidence.

EPA AirData and AQS are treated differently. These sources are valuable for historical and regulatory context, but they are not used as live current AQI feeds. This preserves the distinction between timeliness and validation. Mecklenburg County Air Quality and NC DEQ are also relevant to the project because they provide local and regional context for Charlotte and North Carolina, even when they are not active machine-readable sources in the prototype.

The system also includes fixture and simulation data. Fixture data is used for deterministic development, testing, offline demos, and controlled scenario evaluation. It is explicitly labeled as sample data and must not be presented as real observed telemetry. Simulated anomalies are also clearly separated from real events. They are controlled research artifacts used to test whether the trust and anomaly logic respond appropriately, not evidence of real-world tampering or sensor failure.

Anomaly Detection Methodology

Breathe uses rule-based anomaly detection because explainability is central to the prototype. A more complex model may be useful in future work, but this version needed logic that could be inspected, tested, and explained. The system evaluates a defined set of data quality signals: stale readings, missing readings, sudden spikes or drops, flatline behavior, disagreement between comparable sensors or sources, limited live history, and source coverage limitations.

A stale reading indicates that the latest available data is too old to support confident interpretation. Missing readings indicate that expected data is absent when the system has enough evidence to infer an expected cadence. Sudden spikes or drops identify large changes between comparable readings, while flatline behavior identifies repeated identical values across enough points to suggest possible reporting issues. Sensor disagreement is used only when

multiple comparable sources or sensors are available. If only one comparable source exists, Breathe treats agreement as limited rather than passing.

The system also separates true anomalies from evidence limitations. This distinction became important during real OpenAQ integration because public telemetry can be sparse. Sparse history should not automatically be treated as a severe current failure. Instead, Breathe labels limited history or source coverage gaps as limitations that reduce confidence without implying that a sensor is broken or that the reading is false.

Data Trust Score Methodology

The data trust score is an explainable 0-100 confidence signal that summarizes the quality of the available telemetry evidence. It begins from a high-confidence baseline and applies penalties when the system detects data quality concerns. The score considers freshness, completeness, volatility, continuity, agreement, and evidence availability. The final score is mapped into a trust category such as High Trust, Medium Trust, Low Trust, or Unreliable.

The trust score is intentionally not a health or safety score. A high trust score does not mean the air is safe, and a low trust score does not mean the air is dangerous. Instead, the score communicates whether the data behind the air quality reading appears recent, complete, consistent, and supported by enough evidence. This allows Breathe to separate two questions that are often blended in public dashboards: what the air quality appears to be, and how much confidence the user should place in the data behind that result.

The scoring approach is conservative. It avoids giving perfect confidence when real-time data is thin, but it also avoids over-penalizing sparse public telemetry as if it were automatically a current sensor failure. This design choice keeps the score useful without overstating what the system can know.

User Interface Design: Everyday Use and Research Transparency

Breathe separates its interface into everyday-facing and research-facing layers. The everyday-facing layer includes Today, Act, Trust, and Learn views that help users interpret current air quality, understand source confidence, and connect readings to cautious action without being overwhelmed by implementation details. These views present the current AQI or estimated AQI, category, primary pollutant, source status, trust score, and plain-language explanations. The goal is to make uncertainty visible without turning the interface into a technical report.

Research Mode exposes the evidence behind the interface. It provides source audit details, normalized readings, anomaly explanations, simulation controls, trust scoring logic, and report or trace exports. This mode is intended for technical users, reviewers, researchers, or anyone who wants to inspect how Breathe reached its interpretation.

This split matters because trust communication can easily become too complex. If all technical details are shown at once, everyday users may ignore them or become confused. If technical details are hidden completely, the dashboard becomes another black box. Breathe uses everyday-facing views and Research Mode to balance clarity with transparency. Because the project is a prototype, evaluation focuses on implementation correctness, reproducibility, and claim discipline rather than user-impact claims.

Controlled Evaluation Design

Breathe's controlled evaluation is designed to test whether the anomaly detection and trust scoring logic responds correctly to known data quality problems. Because the project should not claim real-world tampering or field validation without evidence, the evaluation uses

controlled scenarios rather than unverified real incidents. These scenarios allow the system to be tested against expected conditions while preserving research honesty.

The controlled scenarios include a normal baseline, missing readings, stale readings, sudden PM2.5 or AQI spikes, sudden drops, flatline behavior, sensor disagreement, replay-like behavior, and fake-normal readings. Each scenario has an expected anomaly result. The evaluation runner compares the expected result against the detected result and reports true positives, false positives, false negatives, precision, and recall.

This evaluation design is useful because it tests the logic of the system in a reproducible way. However, it is not the same as real-world validation. The scenarios show that Breathe can identify controlled data quality patterns, but they do not prove that the system can detect every real-world sensor issue, environmental event, or source failure. Future evaluation should use real observational datasets with clearly labeled injected anomalies and, eventually, user studies to evaluate whether the trust explanations improve public understanding.

Prototype Validation / Results

The current Breathe prototype validates the feasibility of the system design across implementation, testing, evaluation, and live deployment. The application includes a React/Vite frontend, a FastAPI backend, OpenAQ live public telemetry ingestion, optional AirNow support, EPA AirData/AQS as a historical reference path, fixture fallback, source provenance, anomaly detection, trust scoring, simulations, reports, and controlled evaluation tooling. The interface includes Today, Act, Trust, Learn, and Research views, allowing the system to support both everyday interpretation and technical review. The live prototype is available at <https://breatheair.duckdns.org>.

In the current verified state, the backend test suite passed with 59 tests, Python compile checks passed, the frontend production build completed successfully, and the controlled evaluation runner passed strict checks. The controlled evaluation reported 12 true positives, 0 false positives, 0 false negatives, 1.0 precision, and 1.0 recall across the implemented controlled scenarios. These results show that the rule-based anomaly logic performs correctly against the current scenario set. They should be interpreted as controlled regression results, not as field validation.

The deployed system was also smoke-tested through its public HTTPS endpoint. The live site returned a successful HTTPS response, HTTP redirected to HTTPS, the Open Graph image was available publicly, and the API health endpoint responded successfully. Live API checks confirmed that Breathe resolved the active source mode to OpenAQ, classified the data as real-time preliminary telemetry, avoided fixture fallback, labeled the AQI as estimated when derived from pollutant readings, and returned trust and anomaly evidence without exposing API keys. The live Charlotte response produced a High Trust score with evidence limitations related to sparse recent source history, rather than active operational anomalies.

AirNow was also verified as an optional explicit source path through a live readings endpoint, but OpenAQ remains the primary active source in the default deployed flow. EPA AirData/AQS remains historical/reference-only. These validation results demonstrate that Breathe can combine public telemetry, source classification, normalized data modeling, anomaly detection, explainable trust scoring, and research snapshot generation in one working system. The most important result is not that any single AQI value was correct, but that the system can explain where the data came from, how it was interpreted, and what limitations should be considered.

Limitations

Breathe has several important limitations, and they are part of the research contribution rather than an afterthought. First, Breathe is not an official air quality authority and should not replace the EPA, AirNow, OpenAQ, Mecklenburg County Air Quality, NC DEQ, or public health guidance. It is a research prototype that explores how data trust can be communicated, not a regulatory or medical decision tool.

Second, the trust score is limited by the data available to the system. Real-time public telemetry can be sparse, delayed, preliminary, or uneven across locations. When only limited evidence is available, Breathe can communicate that limitation, but it cannot create certainty where the data does not support it. Similarly, an estimated AQI derived from pollutant readings should not be treated the same as a source-provided or regulatory AQI value.

Third, the controlled evaluation is not field validation. The current scenarios test known patterns such as missing data, stale readings, spikes, drops, flatlines, disagreement, and replay-like behavior. This is useful for verifying system behavior, but it does not prove that Breathe can detect all real-world environmental data problems. Real-world validation would require broader datasets, comparison against official sources, and careful analysis across different locations, pollutants, and time periods.

Fourth, the user-facing design has not yet been evaluated through a formal user study. The interface is designed to be clear and honest, but further work is needed to test whether users actually understand the distinction between air quality risk and data trust. Without that study, Breathe can claim to implement trust-aware design, but it cannot yet claim that the design improves user comprehension or decision-making.

Future Work

Future work should focus on broader evaluation, stronger source comparison, user-centered assessment, and operational hardening. The first technical direction is to expand live-source verification beyond the current Charlotte-first deployment. OpenAQ currently provides the primary live preliminary telemetry path, while AirNow is implemented and can serve as an optional current AQI source when configured. Future versions should compare OpenAQ-derived estimated AQI with AirNow source-provided AQI across more locations and time windows, while continuing to label each source according to its validation status and limitations.

A second direction is broader real-data evaluation. The current controlled scenarios are useful for regression testing because they provide known expected anomaly outcomes, but they are not field validation. Future evaluation should use larger observational datasets, additional cities, multiple pollutants, and longer time periods. Controlled anomalies could be injected into real observational traces while remaining clearly labeled as experimental manipulations. This would allow Breathe to test anomaly detection under more realistic data conditions without claiming that simulated scenarios are real-world events.

A third direction is user-centered evaluation. Breathe should be tested through a small study comparing an AQI-only interface against a trust-aware interface. The study could measure clarity, perceived trust, decision confidence, usefulness, and whether users understand that the trust score refers to data quality rather than air safety. This is important because a trust-aware design is only successful if users interpret the trust layer correctly.

Finally, the deployed system should be hardened operationally. The current AWS EC2 deployment is appropriate for a public research prototype, but it is not a high-availability production system. Future infrastructure work could add persistent source-run storage, report

snapshot persistence, stronger observability, uptime monitoring, automated backups, managed secrets, and eventually a more resilient hosting architecture if usage grows. These improvements would strengthen Breathe as both a research artifact and a public portfolio system while preserving the project's conservative claims about data trust, uncertainty, and source provenance.

Conclusion

Breathe demonstrates how an air quality dashboard can move beyond reporting a single environmental reading and begin communicating the trustworthiness of the data behind that reading. Public dashboards often make air quality easier to understand, but they can also make complex data pipelines appear simpler and more certain than they really are. Breathe responds to that gap by separating air quality risk from data confidence. It asks not only what the current air quality appears to be, but whether the available telemetry appears fresh, complete, consistent, and supported by enough evidence.

The prototype combines real public telemetry, source provenance, normalized data modeling, anomaly detection, explainable trust scoring, controlled simulations, and research snapshot generation. Its purpose is not to replace official environmental systems or declare whether an air quality reading is objectively correct. Instead, Breathe explores how a public-facing interface can make uncertainty visible without overwhelming the user. By labeling preliminary data, estimated AQI values, fixture fallback, and controlled simulations, the system treats data honesty as part of the user experience.

The results of the prototype show that this approach is feasible. Breathe can ingest real public telemetry, preserve source context, evaluate data quality signals, and present both everyday guidance and technical evidence through separate interface modes. The controlled evaluation also shows that the current rule-based anomaly logic responds correctly to the

implemented scenario set. These results do not prove field validity, but they do establish Breathe as a working research prototype for trust-aware environmental telemetry.

Ultimately, Breathe argues that public environmental dashboards should not ask users to accept a number without context. As data systems become more important to everyday decisions, trust should not remain hidden behind the interface. A responsible dashboard should help users understand both what the data says and how much confidence the system has in the data itself.

Significance

For future research, Breathe suggests that trust communication should be treated as a first-class part of public dashboard design. Environmental systems should not only make risk easier to see; they should also communicate uncertainty, data quality, and source limitations clearly. In that sense, Breathe contributes a small but meaningful step toward public technology that is not only informative but also honest about how much confidence users should place in the information being shown.

This matters because environmental data can affect real decisions. People may use air quality information to decide whether to exercise outside, send children outdoors, commute, work outside, or take precautions for respiratory conditions. A trust-aware dashboard does not replace official guidance, but it can help people interpret public information more carefully. By showing source provenance, trust explanations, and data limitations, Breathe supports more informed decision-making without claiming more authority than the data allows.

The project is also significant from a computer science perspective. Breathe is not only a visual dashboard; it is a deployed systems prototype involving source selection, normalized telemetry, rule-based anomaly detection, explainable scoring, simulation, testing, report generation, and cloud deployment through AWS EC2, Nginx, systemd, HTTPS, and GitHub

Actions. This makes it a useful foundation for future work in trustworthy data systems, environmental informatics, and public-interest computing.

For future research, Breathe suggests that trust communication should be treated as a first-class part of public dashboard design. Environmental systems should not only make risk easier to see; they should also communicate uncertainty, data quality, and source limitations clearly. In that sense, Breathe contributes a small but meaningful step toward public technology that is not only informative but also honest about how much confidence users should place in the information being shown.

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